

A more capable cap, with a base trained to support it

REPORT 2 / CONCISE RESEARCH PROPOSAL

Recommendation based on the reviewed results. The design below is proposed work, not a measured improvement.

Recommendation. Give the cap a learned ability to recognize, interpret and apply new knowledge. Train the base and cap together on short learning episodes, rewarding successful updates and preservation of earlier knowledge.

Why change both?

The present cap stores a fixed activation correction behind a similarity gate. It can memorize an answer but often fails to recognize a reworded question, or interferes with another memory. Some retention losses occur while memory is not full. More expressive reading and correction mechanisms directly address these failures. [P1]

The distillation starts with a copy of the teacher and trains it to match that teacher. In ideal arithmetic, the copy already minimizes the pure output-matching loss. This explains why close fidelity alone gives little reason to expect more editable internal representations. The new objective must explicitly reward being easy to update. [P2]

The proposed cap

Capability	Simple implementation idea
Recognize when a memory applies	Learn a reader from several transformer depths and the question context. Train with paraphrases and similar-looking questions that should remain unchanged.
Apply knowledge in context	Store an encoded fact or correction. A small neural network combines it with the current question to generate the appropriate activation changes.
Coordinate the changes	Use one shared internal state to generate corrections at several depths together.
Remember quickly; generalize gradually	Write new experiences to episodic memory. Use a slower, replay-supported learning process to acquire reusable patterns.

A practical starting scale is a few million trainable cap parameters, two or three small layers and a 256-512-dimensional internal state, plus bounded memory. These are pilot choices, not established optimal settings.

Example: after learning "Mira now works at Helix Labs," answer "Who employs Mira?" correctly while preserving the answer to "Where did Mira previously work?" The missing capability is recognizing the new fact's scope and using it appropriately.

Train for the ability to learn

Distillation should preserve competence and develop an interface that supports useful updates.

Each training episode simulates a small piece of continual learning. The system starts with earlier knowledge, receives a new fact or rule, updates the cap, and is tested on new applications plus old and unrelated examples.

Training step	What it teaches
1. Preserve ordinary behavior	Keep the cap-disabled base close to the teacher on ordinary text.
2. Teach a new fact or rule	Allow only a small, declared number of cap updates using the teaching examples.
3. Test new formulations	Reward correct answers to unseen paraphrases and applications of the taught information.
4. Recheck earlier knowledge	Penalize damage to old corrections and closely related questions outside the new fact's scope.
5. Improve the learning system	Use the episode's results to improve the cap and selected base parameters, while measuring the required computation.

Teacher fidelity applies to ordinary base behavior; revised questions are scored against the new information. This avoids asking the edited system to reproduce an obsolete teacher answer. Final evaluation must use facts and entities excluded from training.

Where predictive coding can contribute

A subsequent cap version can maintain a small latent state that predicts and reconciles observations from the tapped transformer depths, conditioned on retrieved memories. A few internal adjustment steps could produce coordinated corrections. Initially, the observations can remain fixed while the small cap settles, followed by one corrected base pass.

This gives iterative computation a useful role inside the cap. It is a hypothesis: the comparison must show that the extra adjustment improves answers enough to justify its cost. Prediction uses the prompt and stored knowledge; the correct answer is supplied only during learning.

Keep the learning claims precise

First build a differentiable reference that learns the episodic objective reliably. An ePC-capable base trained by ordinary backpropagation is a hybrid system. A separate condition must test explicit ePC updates before claiming an advantage from that learning rule.

During each evaluation stream, hold the base fixed so cap learning can be assessed cleanly. A later, separate experiment can allow slow local consolidation into the backbone; that is needed to test the proposed advantage of local backbone plasticity.

A small experiment that can decide

Start with clear failure diagnoses, then test the coordinated change.

First diagnostic: give the current cap the correct stored entry on paraphrase questions. A large improvement points to retrieval failure; persistent failure points to inadequate correction expressiveness. Use this oracle only for diagnosis, never as a deployable score.

The main comparison

Base	Current cap	More capable cap
Current distilled ePC	Existing reference	Does the cap alone help?
Base trained for episodic editability	Does the base change alone help?	Does coordinated training help most?

For this table, use the same cap-update method across cells. Test predictive-coding learning and cap settling as additional controlled switches. Include the original BP base and a BP-trained counterpart of the strongest revised system with comparable data, objectives and compute.

What would count as progress?

- More correct answers to new paraphrases and applications after a long stream of edits.
- Strong initial acquisition and retained original answers, with little damage to earlier or unrelated knowledge.
- A useful cost trade-off after counting cap pretraining, base preparation, learning, memory and prediction.

Start on short development streams, then extend to 1,000 edits and a small controlled task that requires combining two learned facts. Use separate final test data and fixed decision rules. A result that helps equally on BP is still useful; it would support the cap and objective without establishing an ePC-specific advantage.

Recommended implementation order

Learned retrieval and context-dependent corrections first; episodic base/cap training second; cap-level predictive-coding inference third. This sequence produces interpretable intermediate results while pursuing the coordinated design. Preserve the current completed experiment as the baseline.

Sources and related work

- P1** Observed retrieval, retention and capacity findings. [docs/report.md](#).
- P2** Teacher-copy initialization; `pc/kd_energy.py` defines teacher matching. [src/pccap/distill/train.py](#).
- P3** Mitchell et al. (2022), Memory-Based Model Editing at Scale (SERAC): learned interpretation and scope of stored edits.
- P4** Chen et al., Lifelong Knowledge Editing with Retrieval-Augmented Continuous Prompt Learning (RECIPE): jointly trained retrieval and knowledge application.

The related papers motivate learned memory interpretation; they do not validate the proposed ePC design. The companion coding-agent report specifies interfaces, implementation stages and evaluation gates.